

Lecture 4: CIFAR-10, CNNs, Regularisation
Class Activation Maps, Adversarial Attacks
2D Ising Model, T_c from a CNN, the Chinese Room

Deep Learning in Classical and Quantum Physics

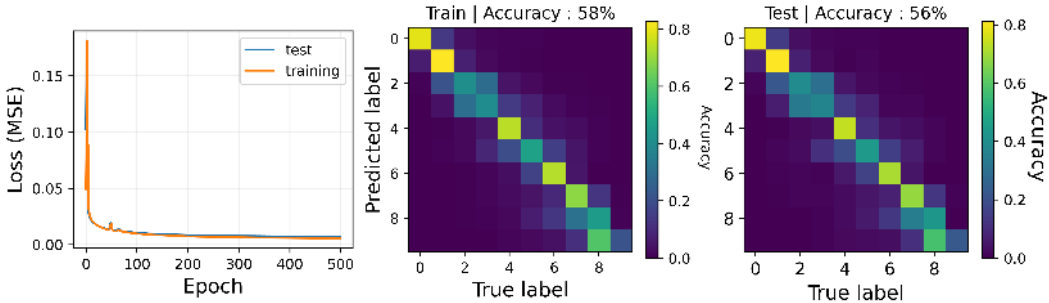
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Recap: MNIST (Modified National Institute of Standards and Technology) Digits

dense, 1 output, MSE (mean squared error)

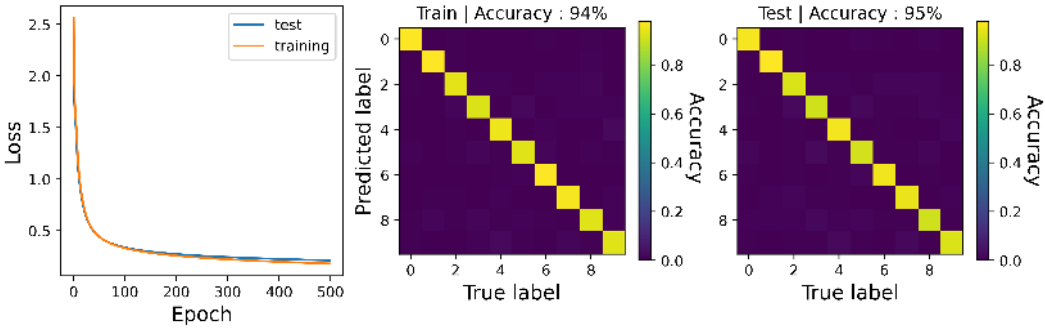
train 58%, test 56%



Ten Outputs with Cross-Entropy Classify the Digits

dense, 10 outputs, cross-entropy

train 94%, test 95%



The formulation of the problem matters as much as the network.

Today: the same recipe on a harder dataset.

1

Hello World #2: CIFAR-10 and Convolutional Neural Networks (CNNs)

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CIFAR-10 Dataset (Canadian Institute For Advanced Research)



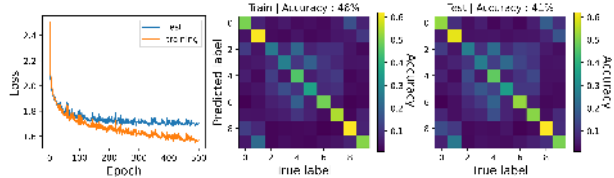
- 10 classes: aeroplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck
- 50 000 training, 10 000 test images
- 32×32 pixels in **3 colour channels** (R, G, B): each image is a $32 \times 32 \times 3$ tensor
- MNIST was $28 \times 28 \times 1$

A. Krizhevsky, *Learning Multiple Layers of Features from Tiny Images* (2009), cs.toronto.edu/~kriz/cifar.html

CIFAR-10: The Same Naive Network + Cross-Entropy

Exactly the MNIST recipe:

- **flatten** the image:
  $32 \cdot 32 \cdot 3 = 3072$ inputs
- one hidden layer, 21 sigmoid nodes
- 10 outputs, softmax, cross-entropy



$3072 \cdot 21 + 21 + 21 \cdot 10 + 10 = 64\,753$

parameters

- **Poor performance:** about 46% training, 41% test accuracy — the model *underfits*

What do we lack in our approach?

Flattening Loses Spatial Structure

Neighbouring pixels are correlated: an edge, a texture, an eye are *local* patterns of a few nearby pixels.

After flattening, the dense layer treats pixel #1 and pixel #2 as unrelated as pixel #1 and pixel #3000.

Shuffle all pixels with one fixed permutation: a dense network learns equally well — it never used the 2D layout.

What an image model should know

1. **Locality:** look at small neighbourhoods
2. **Weight sharing:** the same edge detector is useful everywhere in the image
3. **Translation:** a cat in the corner is still a cat

$\Rightarrow$ **capture spatial correlations** with *convolutions*.

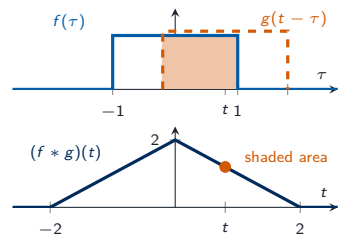
Convolution: the Continuous Definition (1D)

Two functions $f(\tau)$ (the signal) and $g(\tau)$ (the kernel):

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)\, g(t-\tau)\, d\tau$$

Four steps for each output point t:

1. **flip** the kernel: $g(\tau) \to g(-\tau)$
2. **shift** it by t: $g(t-\tau)$
3. **multiply** with the signal: $f(\tau)\, g(t-\tau)$
4. **integrate** over τ: one number, $(f * g)(t)$



Example: two unit boxes on $[-1, 1]$. The overlap of the box with the shifted box has length $2 - |t|$, so $(f * g)(t) = \max(0,\, 2 - |t|)$: a **triangle**.

In physics: a measured spectrum = true spectrum $*$ instrument response; a blurred photo = sharp image $*$ point-spread function.

Convolution: the Continuous Definition (2D)

For functions of two variables, e.g. an image $f(x, y)$ and a kernel $g(x, y)$:

$$(f * g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u, v)\, g(x - u,\ y - v)\, du\, dv$$

The same four steps: flip g in both directions, shift it to (x, y), multiply, integrate over the plane.

Example: $g =$ a 2D Gaussian $\Rightarrow$ $f * g$ is f **blurred**: every point replaced by a weighted average of its neighbourhood.

Convolution Is Linear, Commutative and Shift-Equivariant

Properties (both in 1D and 2D)

- **linear:** $f * (ag_1 + bg_2) = a\, f * g_1 + b\, f * g_2$
- **commutative:** $f * g = g * f$ (substitute $\tau' = t - \tau$)
- **shift-equivariant:** shifting the input shifts the output by the same amount, $f(\cdot - a) * g = (f * g)(\cdot - a)$
- **convolution theorem:** $\mathcal{F}[f * g] = \mathcal{F}[f] \cdot \mathcal{F}[g]$ — in Fourier space a convolution is a product

Discrete Convolution in 1D

Signal sampled on a grid, $x[n]$; kernel with a few entries, $g[m]$. The integral becomes a sum:

$$(x * g)[n] = \sum_m g[m]\, x[n - m]$$

Example

$x = (1,\ 2,\ 4,\ 7,\ 7,\ 7), \quad g = (g[0], g[1], g[2]) = (1,\ 0,\ -1)$
$(x * g)[n] = x[n] - x[n-2]$ (only positions where the kernel fits fully):
$4 - 1 = 3, \quad 7 - 2 = 5, \quad 7 - 4 = 3, \quad 7 - 7 = 0 \quad \Rightarrow \quad x * g = (3,\ 5,\ 3,\ 0)$

- kernel $(1, 0, -1)$: a **difference** — large where the signal changes, 0 where it is flat
- kernel $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$: a **moving average** — $x * g = (2.33,\ 4.33,\ 6,\ 7)$, a smoothed signal

Shift-equivariance: prepend two zeros, $x' = (0, 0, 1, 2, 4, 7, 7, 7)$:
$x' * g = (1,\ 2,\ 3,\ 5,\ 3,\ 0)$ — the same pattern $(3, 5, 3, 0)$, moved by two places

Discrete Convolution in 2D

Image $x(i,j)$, kernel $G(m,n)$ of size $k \times k$:

$$\text{convolution:} \quad (G * x)(i,j) = \sum_m \sum_n G(m,n)\, x(i-m,\, j-n)$$

cross-correlation (what CNN layers compute): $\quad S(i,j) = \sum_m \sum_n G(m,n)\, x(i+m,\, j+n)$

The two differ only by flipping the kernel. The kernel is *learned*, so the flip does not matter: deep-learning libraries compute the cross-correlation and still call it "convolution".

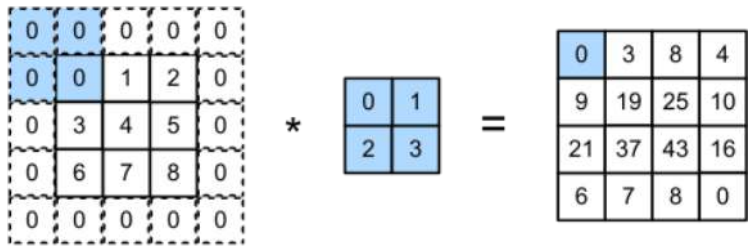
With channels c: $S(i,j) = \sum_c \sum_{m,n} G(m,n,c)\, x(i+m, j+n, c)$; S is a **feature map**.

A Two-by-Two Kernel Detects a Vertical Edge

Example: a vertical edge

$x = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$, $G = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$, $S = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ (true convolution: -2 instead of 2)

e.g. middle column: patch $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$, $-0 + 1 - 0 + 1 = 2$: the kernel straddles the edge



2×2 kernel on a zero-padded 3×3 input: 4×4 output

The Effect of a Convolution

- **a local weighted sum:** each output looks only at a $k \times k$ neighbourhood
- **the same weights everywhere** (weight sharing): k^2 parameters, whatever the image size
- **a pattern detector:** $S(i, j)$ is the scalar product of the kernel with the patch at (i, j), so it is largest where the patch *looks like* the kernel
- **shift-equivariant:** move the object, the feature map moves with it

In a CNN the kernels are not designed by hand — they are learned by gradient descent.

The Kernel Decides the Operation

Kernel $\rightarrow$ operation	
all entries equal	blur (average)
Gaussian	smooth blur
$(-1, 1)$	derivative, edges
$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	discrete Laplacian ∇^2
centre $+$, neighbours $-$	sharpen

Physics: the Laplacian kernel is the finite-difference ∇^2 used in the heat and Schrödinger equations.

Kernels Detect Features: Sobel Filters

$$\vec{G}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \qquad \vec{G}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

approximate the intensity gradient: they light up where the image changes sharply — **edges**



original · vertical edges ($\vec{G}_x$) · horizontal edges ($\vec{G}_y$)

Different Kernels, Different Effects

Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	
Ridge or edge detection	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
Box blur (normalized)	$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	
Gaussian blur 3 × 3 (approximation)	$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	

Identity, edge detection, sharpening, blurring — all are 3×3 kernels.

Idea of a CNN: do not design the kernels — learn them.

Kernels Are Learnable Parameters

-1	0	$+1$
-2	0	$+2$
-1	0	$+1$

Sobel $\vec{G}_x$:
fixed, designed by hand

G_{11}	G_{12}	G_{13}
G_{21}	G_{22}	G_{23}
G_{31}	G_{32}	G_{33}

CNN kernel:
9 trainable weights

- the entries $G(m, n, c)$ and the bias b are the **weights** of a convolutional layer, as W and b are for a dense layer
- they start random and are trained by gradient descent on the same loss:

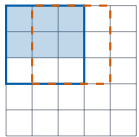
$$G(m, n, c) \leftarrow G(m, n, c) - \eta \frac{\partial \mathcal{L}}{\partial G(m, n, c)}$$

- backpropagation gives the gradient; one kernel is shared by all positions, so its gradient sums over all of them

After training, first-layer kernels often look like edge detectors.

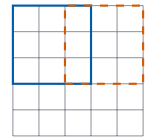
Kernel Size, Stride, Padding

stride 1



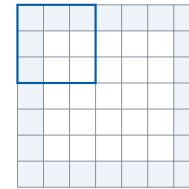
5×5 input, 3×3 kernel $\rightarrow 3 \times 3$ output

stride 2



the kernel jumps 2 pixels $\rightarrow 2 \times 2$ output

padding 1



zeros around the border $\rightarrow 5 \times 5$ output ("same")

Output size along one dimension

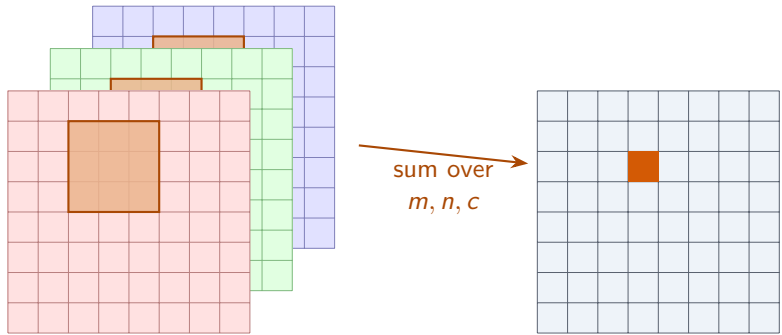
$$O = \left\lfloor \frac{W - k + 2P}{s} \right\rfloor + 1$$

W : input size, k : kernel size
P : padding, s : stride

CIFAR-10: $W = 32$, $k = 3$, $P = 1$, $s = 1 \Rightarrow O = 32$; with $s = 2$: $O = 16$

Images with Three Channels

A colour image has three numbers per pixel, red, green and blue: $x(i, j, c)$ with $c \in \{R, G, B\}$. A CIFAR-10 image is $32 \times 32 \times 3$.



input $32 \times 32 \times 3$: red, green, blue layers;
kernel $3 \times 3 \times 3$ (orange)

one feature map
(one channel)

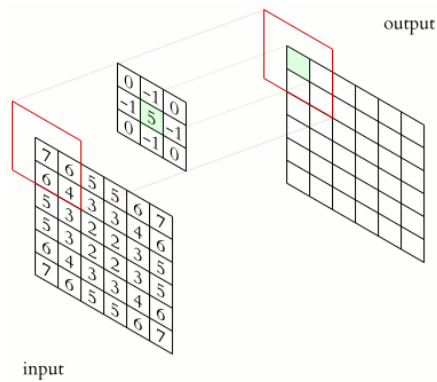
One Kernel Sums over All Input Channels

$$S(i,j \mid c) = \sum_{m,n} G(m,n,c)\, x(i+m,\, j+n,\, c), \qquad S(i,j) = \sum_{c} S(i,j \mid c) + b$$

Each channel is convolved with its own slice of the kernel, and the three results are added: one kernel gives *one* feature map; C_{out} kernels give C_{out} feature maps, which are the channels of the next layer's input.

Channels and Feature Maps

- input with C_{in} channels (RGB: $C_{\text{in}} = 3$) $\rightarrow$ each kernel has shape $k \times k \times C_{\text{in}}$
- C_{out} kernels $\rightarrow$ C_{out} feature maps (output channels)
- **the kernel entries are the trainable parameters**



A Convolutional Layer Has Few Parameters

Parameter count, first layer on CIFAR-10

Conv2D, 32 kernels 3×3: $3 \cdot 3 \cdot 3 \cdot 32 + 32 = 896$

Dense, 21 nodes on the flattened image: $3072 \cdot 21 + 21 = 64\,533$

Fewer parameters, because one kernel is *shared* across all positions.

Pooling: Max, Average, Global Average

Window 2×2, stride 2: each 2×2 block of the feature map becomes *one* number.

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 7 & 8 \\ 4 & 2 & 1 & 3 \\ 9 & 8 & 6 & 5 \end{bmatrix}$$

$$\xrightarrow{\text{max pooling}} \begin{bmatrix} 6 & 8 \\ 9 & 6 \end{bmatrix}$$

$$\xrightarrow{\text{average pooling}} \begin{bmatrix} 3.75 & 5.25 \\ 5.75 & 3.75 \end{bmatrix}$$

$$\xrightarrow{\text{global average pooling}} 4.625$$

top-left block: $\max(1,3,5,6) = 6$, $(1+3+5+6)/4 = 3.75$; global average: sum of all 16 numbers $= 74$, $74/16 = 4.625$

Three Kinds of Pooling

Max pooling

keeps the *strongest* response in each window: "is the feature present here?" — the usual choice between conv layers

Average pooling

keeps the *mean* response of each window: a smoothed summary

Global average pooling

the window is the *whole* map: one number per feature map; used at the end of a network

No trainable parameters in any of them. A learned alternative: a convolution with stride 2.

Downsampling: Four Reasons

1. Computational cost

one 2×2 pooling: 4 times fewer numbers to store and to process in every later layer
32 maps of 32×32: 32 768 numbers $\rightarrow$ 32 maps of 16×16: 8 192

2. A larger field of view

after pooling, a 3×3 kernel covers a 6×6 region of the map before pooling: deeper layers see larger parts of the image

3. Robustness to small shifts

a feature moved by one pixel usually stays in the same pooling window: the output does not change

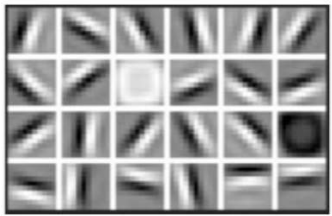
4. Fewer parameters afterwards

the dense layer after "flatten" has one weight per number that enters it: smaller maps, fewer weights, less overfitting

The price: exact positions are lost. Good for "*what* is in the image", a problem when "*where*" matters (segmentation).

CNNs Learn a Hierarchy of Features

- a CNN is a neural network that **learns the kernels** for each channel
- layers of kernels capture a **hierarchy of features**: edges $\rightarrow$ parts $\rightarrow$ objects



first layer



middle layer



deep layer

Second Attempt at CIFAR-10 Classification

First attempt: a dense network

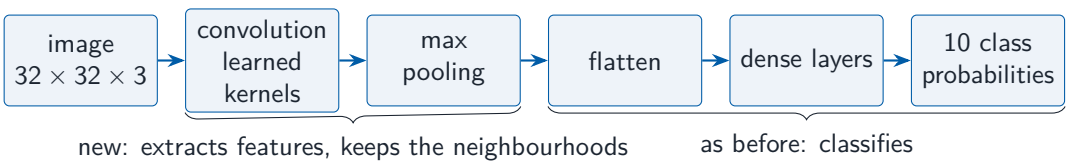
- the image is flattened into 3072 numbers
- which pixels are neighbours is thrown away
- 46% training, 41% test accuracy

Second attempt: a convolutional network

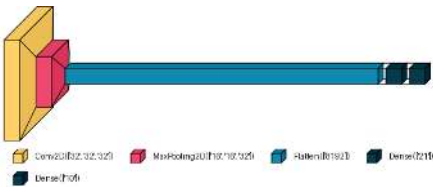
- the image stays an image: $32 \times 32 \times 3$
- kernels look at small neighbourhoods and are *learned*
- pooling reduces the size of the feature maps

Same data, same loss (categorical cross-entropy), same training loop: **only the model changes.**

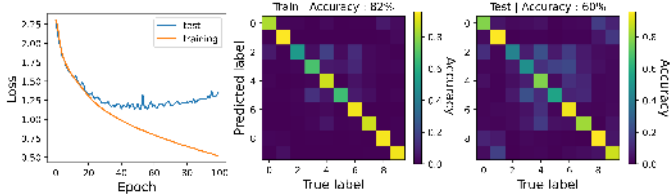
A CNN Extracts Features, Then Classifies



A Small CNN on CIFAR-10

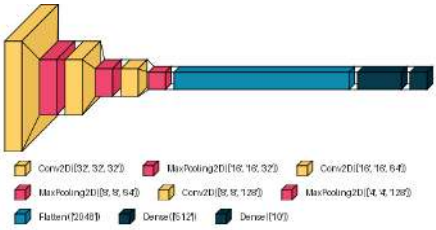


- Conv2D: 32 kernels 3×3, sigmoid, padding "same" $\rightarrow 32 \times 32 \times 32$
- MaxPooling $2 \times 2 \rightarrow 16 \times 16 \times 32$
- flatten $\rightarrow$ 8192, dense 21 (sigmoid), dense 10 (softmax)

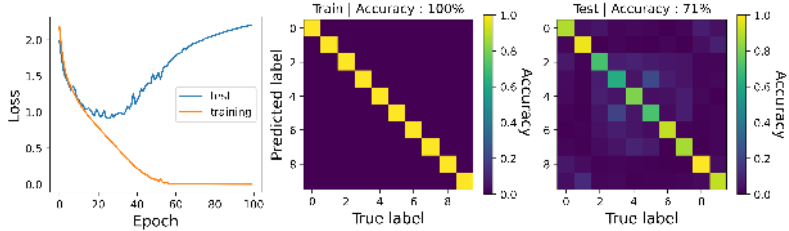


About 82% training, 60% test accuracy — better than 41%, but the test loss **rises after ~40 epochs**: *overfitting*.

More CNN Layers: Still Overfitting



three blocks Conv2D (ReLU = rectified linear unit, $\max(0, x)$) + MaxPooling:
$32 \times 32 \times 32 \to 16 \times 16 \times 64 \to 8 \times 8 \times 128$,
then flatten, dense 512, dense 10



100% training, 71% test accuracy: **more complexity does not cure overfitting**.

2

Regularisation: Curing Overfitting

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Regularisation: Three Common Techniques

Regularisation: any technique that constrains training or model capacity so that the network *generalises* to unseen data instead of memorising the training set.

1. L1 / L2 penalty

a term in the loss that keeps the weights small

2. Dropout

randomly switch off neurons during training

3. Batch normalisation

normalise the inputs of each hidden layer over a mini-batch

Also widely used: *early stopping* (stop when the validation loss starts to rise), *data augmentation* (shifted, flipped, cropped copies of the training images), *learning-rate scheduling*.

L1 / L2 Regularisation

L1 (Lasso)

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{original}} + \lambda \sum_i |\theta_i|$$

drives some weights exactly to zero: *sparse* solutions

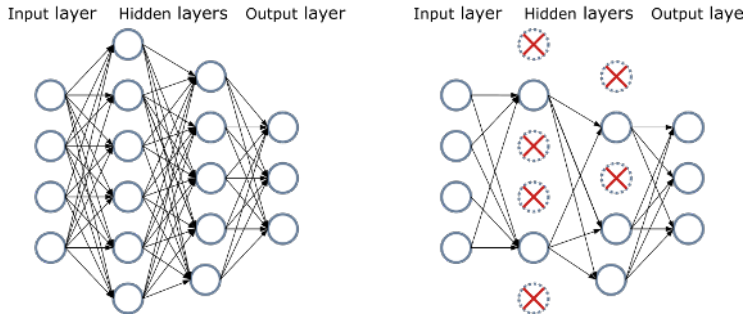
L2 (Ridge, "weight decay")

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{original}} + \lambda \sum_i \theta_i^2$$

discourages large weights: *smoother* models

- θ_i: the weights of the network; λ: the regularisation strength, a hyperparameter
- a **soft constraint**: weights are encouraged, not forced, to stay small — λ plays the role of a Lagrange multiplier
- $\lambda = 0$: no regularisation; λ too large: the network underfits

Dropout



- at every training step each neuron of the layer is switched off with probability p (the *dropout rate*, typically 0.2–0.5)

N. Srivastava et al., *Dropout: A Simple Way to Prevent Neural Networks from Overfitting*, JMLR **15**, 1929 (2014), jmlr.org/papers/v15/srivastava14a.html

Dropout Forces Redundant Representations

- the network cannot rely on any single neuron: it learns **redundant representations**
- each step trains a different sub-network: dropout acts as an **ensemble** of sub-networks
- at test time all neurons are active

Batch Normalisation

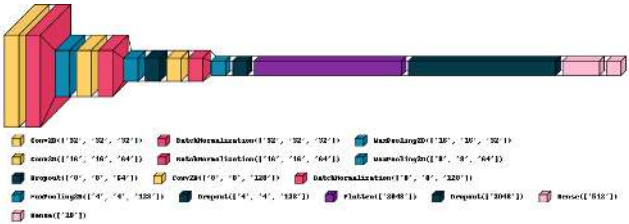
A BatchNorm (batch normalisation) layer normalises the input of a hidden layer over the mini-batch $\{\mathbf{x}_j\}_{j=1}^{N_B}$:

$$\mu_B = \frac{1}{N_B}\sum_{j=1}^{N_B}\mathbf{x}_j, \qquad \sigma_B^2 = \frac{1}{N_B}\sum_{j=1}^{N_B}(\mathbf{x}_j-\mu_B)^2 \qquad \mathbf{x}_j \leftarrow \frac{\mathbf{x}_j-\mu_B}{\sqrt{\sigma_B^2+\epsilon}}, \qquad \mathbf{x}_j \leftarrow \gamma\,\mathbf{x}_j+\beta$$

- step 1: zero mean and unit variance within the mini-batch (ϵ: a small constant for numerical stability)
- step 2: rescale and shift with **trainable** γ and β
- makes the network less dependent on weight initialisation
- reduces sensitivity to weight changes; helps against exploding and vanishing gradients

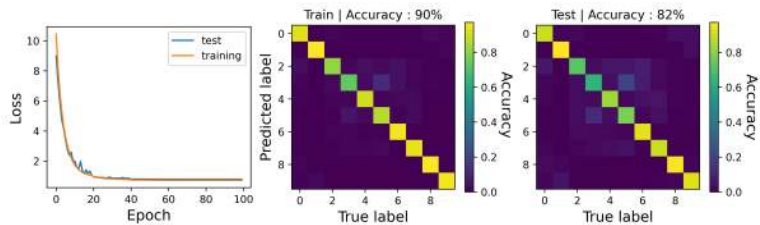
S. Ioffe, C. Szegedy, *Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift*, ICML 2015, arXiv:1502.03167

The Larger CNN with Regularisation



Same three Conv2D blocks, now with batch normalisation and dropout: 90% training, 82% test accuracy.

Before: 100% / 71%.



The training and test losses now stay together.

3

Class Activation Maps: Where the CNN Looks

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Class Activation Maps

input (32 x 32)



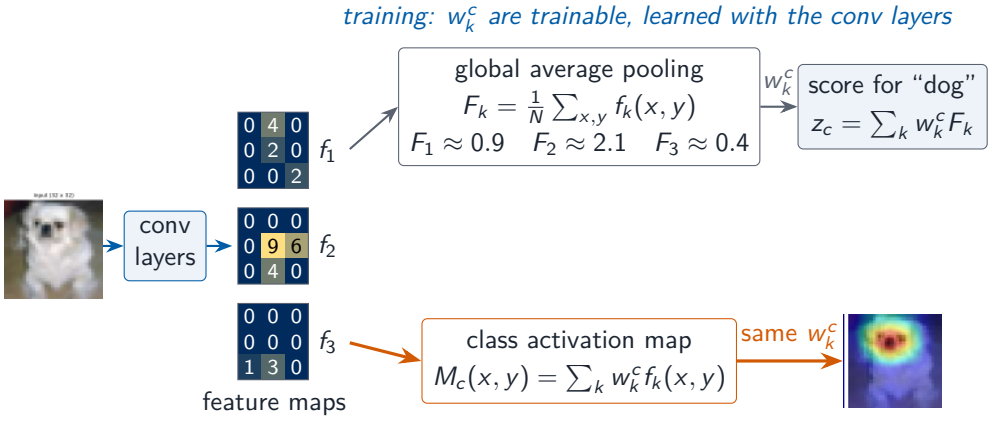
a CIFAR-10 test image; our CNN: "dog", 1.00

Which parts of the image were important for the prediction?

Can we get insight into what the CNN was looking at when it made its prediction?

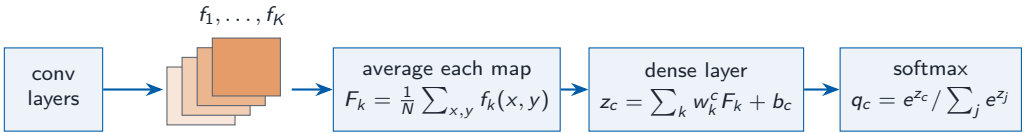
A **class activation map** (CAM) answers with a picture: a map over the image that shows *where* the evidence for a class is.

From Feature Maps to a Class Map



B. Zhou et al., *Learning Deep Features for Discriminative Localization*, CVPR 2016, arXiv:1512.04150

CAM: From the Class Score Back to a Map



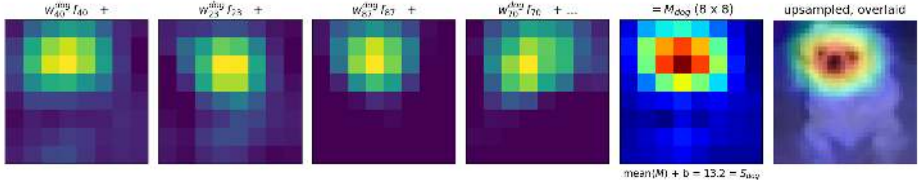
$$z_c = \sum_k w_k^c F_k + b_c = \sum_k w_k^c \frac{1}{N} \sum_{x,y} f_k(x,y) + b_c = \frac{1}{N} \sum_{x,y} \underbrace{\sum_k w_k^c f_k(x,y)}_{M_c(x,y)} + b_c$$

Class activation map

$$M_c(x,y) = \sum_k w_k^c f_k(x,y)$$

the class score is its average over the positions

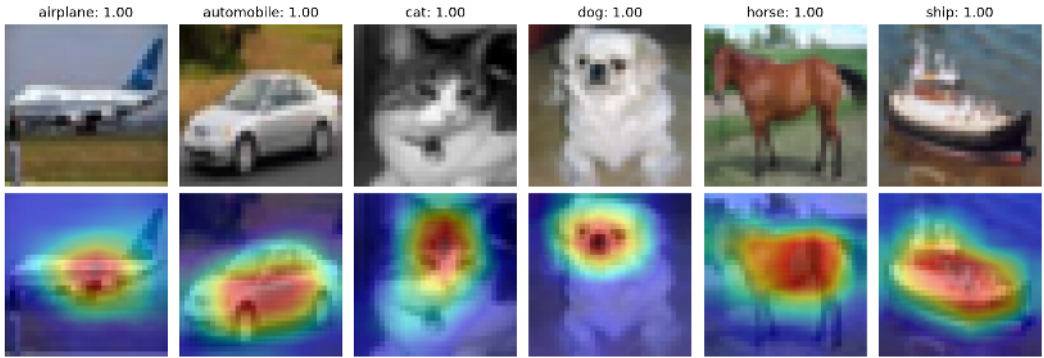
CAM on a Real Network



our CIFAR-10 CNN with GAP: 6 conv layers $\rightarrow$ 128 feature maps of 8×8 $\rightarrow$ GAP $\rightarrow$ 10 outputs, 87.6% test accuracy

top: feature maps times their weights for "dog", added to one 8×8 map; bottom: the map enlarged and laid over the image.

CAM on CIFAR-10 Test Images



Correctly classified test images (top) and their class activation maps (bottom): the evidence sits on the object, not on the background.

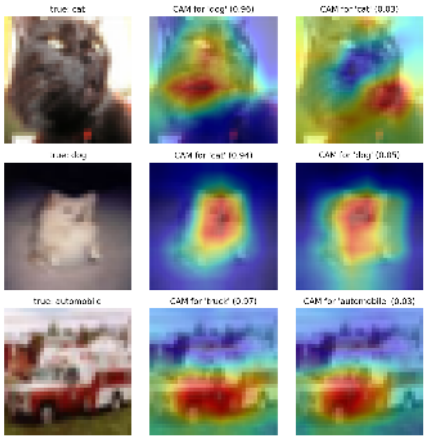
CAM for Two Classes in One Photograph



- the map of a dog class lies on the dog; the map of a cat class lies on the cat
- network: ResNet-18 trained on ImageNet (1000 classes), also ending with global average pooling
- p: softmax probability of that class. It is small: two small animals, 1000 classes to share the probability.

photograph: Socks and Buddy, U.S. National Archives, public domain, Wikimedia Commons

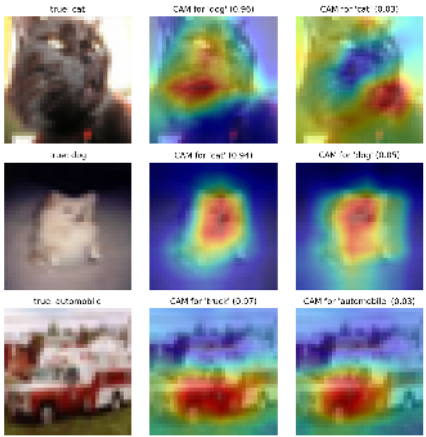
CAM: Looking at Mistakes



Three misclassified test images; the map for the *predicted* class (middle) and for the *true* class (right).

- **cat → "dog" (0.96):** the "dog" evidence sits on the lower part of the face; the weak "cat" evidence is elsewhere. A dark close-up, the outline of the animal is not visible.
- **dog → "cat" (0.94):** both maps cover the animal. The right place, the wrong one of two similar classes.
- **automobile → "truck" (0.97):** both maps on the body and the wheels: the right place, a neighbouring class.

Similar Classes, Few Pixels and Softmax Explain the Mistakes



- similar classes share most features (fur, four legs; wheels, body): their scores differ only by small weights
- 32×32 pixels: the details we would use are not in the image
- softmax must choose one of the 10 classes, and does so with high confidence even when it is wrong

CAM: Uses and Limits

Tells us

- *where* the evidence for a class is
- whether the model looks at the object or at the background (shortcut learning)
- for a mistake: compare the maps of the predicted and the true class

Does not tell us

- *why*: which property of the region mattered
- fine detail: the resolution is that of the last feature map (8×8 here), then upsampled
- anything for networks that do not end in GAP $\rightarrow$ dense

High Accuracy Can Rest on the Wrong Evidence

Be careful: accurate for the wrong reason

a model can reach high accuracy by using the wrong evidence, e.g. telling wolves from huskies by the *snow* in the background.

M. T. Ribeiro et al., *"Why Should I Trust You?": Explaining the Predictions of Any Classifier*, KDD 2016, arXiv:1602.04938

Grad-CAM (Gradient-weighted Class Activation Map) for Any CNN

- $f_k(x, y)$: feature maps of the last convolutional layer; z_c: score of class c before softmax
- CAM needs one weight w_k^c per map; most CNNs have several layers after the last convolution, so there is no such weight
- Grad-CAM uses instead the gradient of the class score with respect to each map, averaged over the N pixels and computed by backpropagation:

$$\alpha_k^c = \frac{1}{N} \sum_{x,y} \frac{\partial z_c}{\partial f_k(x, y)}$$

Grad-CAM Reduces to CAM for a Network Ending in GAP

CAM

$$M_c(x,y) = \sum_k w_k^c \, f_k(x,y)$$

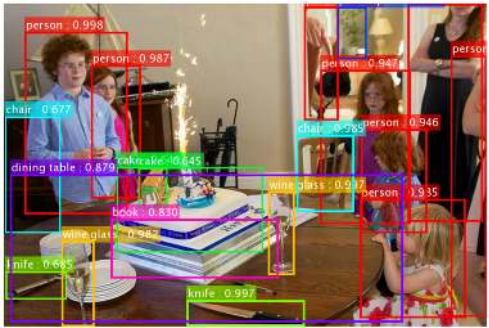
Grad-CAM

$$M_c(x,y) = \mathrm{ReLU}\Big(\sum_k \alpha_k^c \, f_k(x,y) \Big)$$

- ReLU sets negative values to zero: the map shows only the regions that *increase* z_c
- for a GAP network $z_c = \sum_k w_k^c F_k + b_c$, so $\alpha_k^c = w_k^c / N$: Grad-CAM gives the CAM map, up to the factor $1/N$ and the ReLU

R. R. Selvaraju et al., *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*, ICCV 2017, arXiv:1610.02391

Beyond Classification: Detection and Segmentation





Object detection: a box and a class probability for every object in the image

Segmentation and dense prediction: a class (or depth, or edge) for *every pixel*

All built on the same convolutional feature extractor.

Summary So Far

data	model	train	test
MNIST	dense, 1 output, MSE	58%	56%
MNIST	dense, 10 outputs, cross-entropy	94%	95%
CIFAR-10	dense, 10 outputs, cross-entropy	46%	41%
CIFAR-10	small CNN	82%	60%
CIFAR-10	larger CNN	100%	71%
CIFAR-10	larger CNN with batch norm, dropout	90%	82%
CIFAR-10	CNN with GAP, batch norm, augmentation	91%	88%

first six rows: book Ch. 2; last row: our CAM network

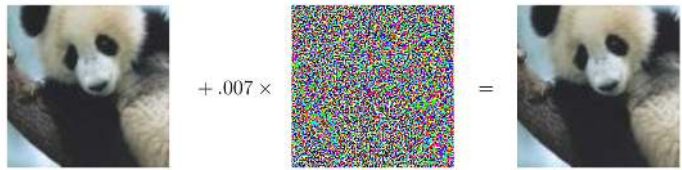
Regularisation closes the gap between training and test.

4

Adversarial Examples and the Failure of Networks

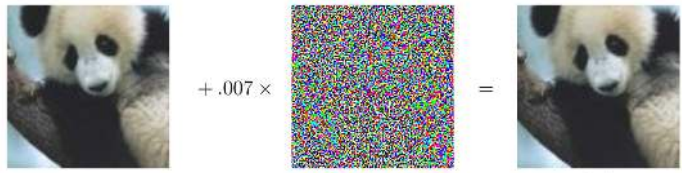
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Adversarial Examples



- **left:** a photograph. The trained classifier says "panda", with 57.7% confidence — correct.
- **middle:** a pattern computed from the network. Shown alone it means nothing: the classifier's best guess is "nematode" (a roundworm), with 8.2%.

Photograph plus Pattern Is Classified as a Gibbon



- **right:** photograph + 0.007× pattern. For us the same photograph; the classifier says "gibbon", with 99.3% confidence.

Such inputs are called **adversarial examples**.

I. J. Goodfellow, J. Shlens, C. Szegedy, *Explaining and Harnessing Adversarial Examples*, ICLR 2015, arXiv:1412.6572

Fast Gradient Sign Method

$$\mathbf{x}_{\text{adv}} = \mathbf{x} + \epsilon \operatorname{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}, y)), \qquad \epsilon = 0.007$$

$J(\theta, \mathbf{x}, y)$	the loss (cross-entropy) of the network with weights θ for the image $\mathbf{x}$ and its true label y
$\nabla_{\mathbf{x}} J$	the gradient with respect to the **image**: one number per pixel value — "if this pixel becomes slightly brighter, how much does the loss grow?"
$\operatorname{sign}(\cdot)$	keep only the direction: $+1$ (brighter raises the loss) or -1 (darker raises the loss)
ϵ	the size of the step, the same for every pixel

Training (what we did so far)

the image is fixed, the *weights* move *down* the loss:

$\theta \leftarrow \theta - \eta \, \nabla_{\theta} J$

Attack

the weights are fixed, the *image* moves *up* the loss:

$\mathbf{x} \leftarrow \mathbf{x} + \epsilon \operatorname{sign}(\nabla_{\mathbf{x}} J)$

The Pattern Is Computed for One Image

Nothing is trained here. There is no second network that generates the pattern. The classifier is already trained and stays untouched; the pattern is *computed* for this one image with one forward and one backward pass — the same backpropagation as in training.

Why such a small step is enough. Each pixel value changes by only ϵ, but *all* of them change, each in its worst direction. For a score that is a weighted sum, $S = \sum_i w_i x_i$, the change is $\epsilon \sum_i |w_i|$: it grows with the number of pixel values, and an image of $224 \times 224 \times 3$ pixels has 150 528 of them.

I. J. Goodfellow, J. Shlens, C. Szegedy, *Explaining and Harnessing Adversarial Examples*, ICLR 2015, arXiv:1412.6572

One-Pixel Attack

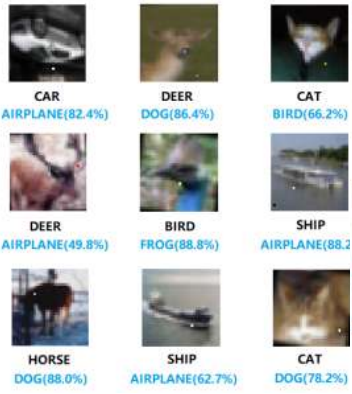


Fig. 1: CIFAR-10



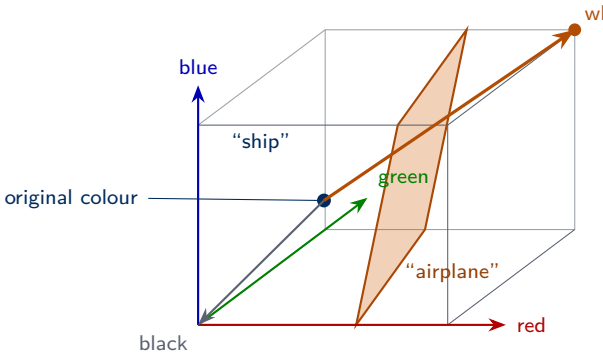
Fig. 2: ImageNet, pixel in the red circle

- black: the true class; blue: the class after changing one pixel, with the network's confidence
- Kaggle CIFAR-10 test set: 67.97% of the images can be moved to at least one other class by one pixel, with 74.03% confidence on average
- ImageNet: 16.04% of the test images, 22.91% confidence

J. Su, D. V. Vargas, K. Sakurai, *One Pixel Attack for Fooling Deep Neural Networks*, IEEE Trans. Evol. Comput. **23**, 828 (2019), doi:10.1109/TEVC.2019.2890858, arXiv:1710.08864

One-Pixel Attack: Mechanism

- an image is one point in a space with 3072 axes
- the network cuts this space into class regions, separated by **decision boundaries**
- one pixel = 3 of these axes: its red, green and blue value
- if a boundary is close along these 3 axes, changing the colour of that pixel moves the image into another class



the cube of all colours of *one* pixel; orange: the decision boundary. To white: the class changes. To black: it does not.

J. Su, D. V. Vargas, K. Sakurai, *One Pixel Attack for Fooling Deep Neural Networks*, IEEE Trans. Evol. Comput. **23**, 828 (2019), doi:10.1109/TEVC.2019.2890858, arXiv:1710.08864

One-Pixel Attack: Finding the Pixel

The search in the paper

- 1024 positions $\times$ 256^3 colours: too many to try all
- a candidate: five numbers (x, y, R, G, B)
- its quality: the probability of the true class — the lower, the better
- differential evolution: 400 random candidates, mix them, keep the better ones, repeat
- black box: no gradients, only output probabilities

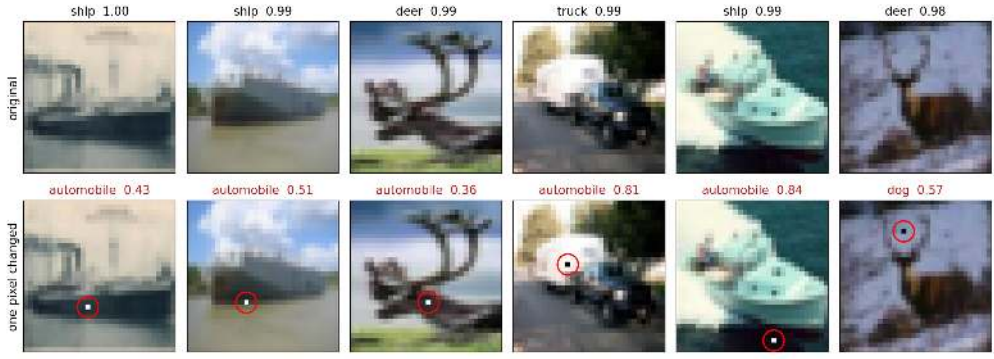
J. Su, D. V. Vargas, K. Sakurai, *One Pixel Attack for Fooling Deep Neural Networks*, IEEE Trans. Evol. Comput. **23**, 828 (2019), doi:10.1109/TEVC.2019.2890858, arXiv:1710.08864

A Random Search on Our CIFAR-10 Network

On our CIFAR-10 network

- the CNN of the CAM section (87.6% test accuracy)
- plain random search: 1000 candidates per image, extreme colours (black, white, . . .)
- 111 of 300 correctly classified images change class (37.0%); average confidence in the wrong class 71%

One-Pixel Attack on Our CIFAR-10 Network



top: original; bottom: one pixel changed (red circle)

Adversarial Stop Sign



left: real graffiti on a stop sign; right: the designed black and white stickers

- a few black and white stickers on a *real* stop sign
- a road-sign classifier reads it as **"Speed Limit 45"**
- 100% of the images taken in the lab, 84.8% of the video frames taken from a moving vehicle
- the perturbation survives changes of distance and viewing angle
- for an autonomous car: a stop sign that is not seen as one

K. Eykholt et al., *Robust Physical-World Attacks on Deep Learning Visual Classification*, CVPR 2018, arXiv:1707.08945

Adversarial T-Shirts



Video frames with two people. The detector (YOLOv2, "You Only Look Once") draws a box around each person it finds; the person wearing the printed pattern gets no box. Attack success rate: 74% in the digital world, 57% with a printed T-shirt on a moving person.

K. Xu et al., *Adversarial T-shirt! Evading Person Detectors in A Physical World*, ECCV 2020, arXiv:1910.11099

Attacks on Deep Networks: Inputs and Training

attack	what the attacker does	reference
at test time: the input is manipulated		
adversarial example	adds a small, designed perturbation to one input	*Intriguing properties of neural networks*, Szegedy et al. 2014
universal perturbation	one pattern that fools the network on most images	*Universal adversarial perturbations*, Moosavi-Dezfooli et al. 2017
adversarial patch	a printed sticker placed anywhere in the scene	*Adversarial Patch*, Brown et al. 2017

Attacks at Training Time Manipulate the Data

attack	what the attacker does	reference
at training time: the data or the training are manipulated		
data poisoning	injects crafted samples into the training set	*Poisoning Attacks against Support Vector Machines*, Biggio et al. 2012
backdoor	hides a trigger: the model is normal until the trigger appears	*BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain*, Gu et al. 2017
attack on federated learning	one participant of a joint training plants a backdoor	*How To Backdoor Federated Learning*, Bagdasaryan et al. 2018

Attacks on Deep Networks: Privacy and Language Models

attack	what the attacker does	reference
against privacy and ownership: information is extracted		
model stealing	rebuilds the model from its answers to queries	*Stealing Machine Learning Models via Prediction APIs*, Tramèr et al. 2016
membership inference	finds out whether a given record was in the training set	*Membership Inference Attacks against Machine Learning Models*, Shokri et al. 2017
model inversion	reconstructs training data from the model's outputs	*Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures*, Fredrikson et al. 2015

Prompt Injection Attacks Language Models

attack	what the attacker does	reference
language models		
prompt injection	hides instructions in the text or web page the model reads	*Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection*, Greshake et al. 2023

Medical records, patient images, unpublished measurements: the privacy attacks concern scientific data directly.

Limits of Deep Networks

About the networks

- high test accuracy says nothing about inputs *outside* the distribution of the training data
- the network does not use the features we use: it uses whatever separates the training classes
- confidence is not correctness: the wrong answers come with high probability

In scientific data, without any attacker

- a new microscope, a different staining, another detector or hospital: the inputs shift, the accuracy can drop
- a network can learn a shortcut: a marker, a scale bar, the background, instead of the object
- tools to check: class activation maps, test data from an *independent* source

5

Optimisation Algorithms: Under the Hood

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Mini-Batch Stochastic Gradient Descent (SGD)

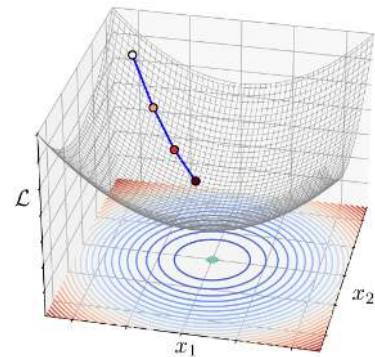
under the hood

Loss on a mini-batch $\mathcal{B}$ of size B:

$$\boldsymbol{g}_t = \nabla_{\boldsymbol{\theta}} \frac{1}{B} \sum_{i \in \mathcal{B}} \ell(f_{\boldsymbol{\theta}}(x_i), y_i)$$

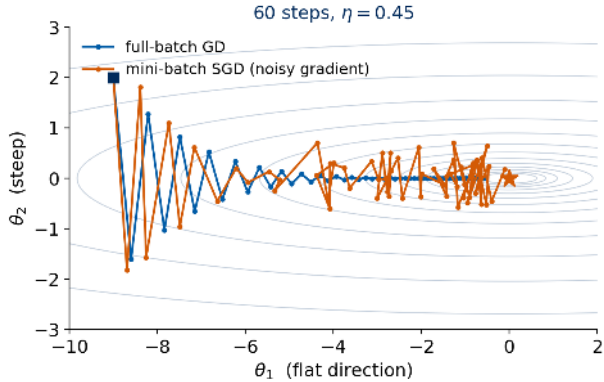
$$\boxed{\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\, \boldsymbol{g}_t}$$

- η: learning rate — the same for every parameter
- noisy gradient: cheap per step, helps escape shallow minima
- slow in flat directions, oscillates in steep ones



SGD on an Elongated Quadratic Loss

under the hood



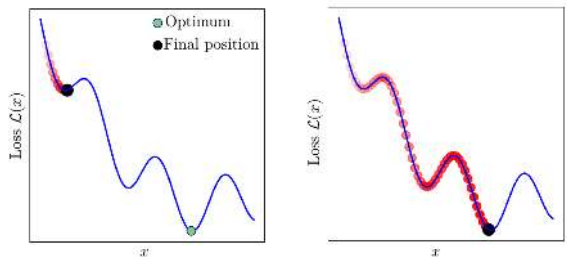
full-batch gradient descent (GD) and mini-batch SGD on the toy loss $\frac{1}{2}(0.1\,\theta_1^2 + 4\,\theta_2^2)$, θ_1 the flat and θ_2 the steep direction

SGD with Momentum

under the hood

$$\boldsymbol{v}_{t+1} = \beta\, \boldsymbol{v}_t + \eta\, \boldsymbol{g}_t, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \boldsymbol{v}_{t+1}$$

- adds a fraction of the previous update into the new one
- accumulates the history: $\boldsymbol{v}_{t+1} = \eta \sum_{k \le t} \beta^{\,t-k} \boldsymbol{g}_k$
- damps oscillations, accelerates along consistent directions
- avoids getting stuck in local minima



left: SGD stops in a local minimum; right: momentum carries it over the barrier to the optimum

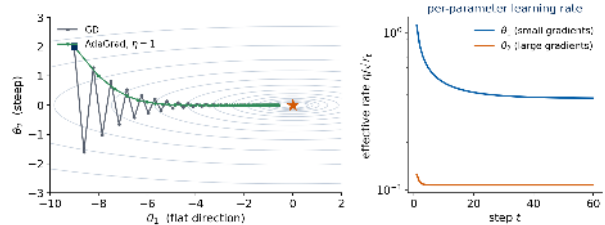
A heavy ball rolling with friction $1 - \beta$.

AdaGrad: Adaptive Gradient Algorithm

under the hood

$$\boldsymbol{r}_{t+1} = \boldsymbol{r}_t + \boldsymbol{g}_t \odot \boldsymbol{g}_t, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\boldsymbol{r}_{t+1}} + \epsilon} \odot \boldsymbol{g}_t$$

- adapts the learning rate **separately for each parameter**
- accumulates squared gradients
- large changes $\rightarrow$ smaller rate; small changes $\rightarrow$ relatively larger rate



Problem: $\boldsymbol{r}_t$ only grows — the rate decays towards zero.

RMSProp (Root Mean Square Propagation) / Adadelta



under the hood

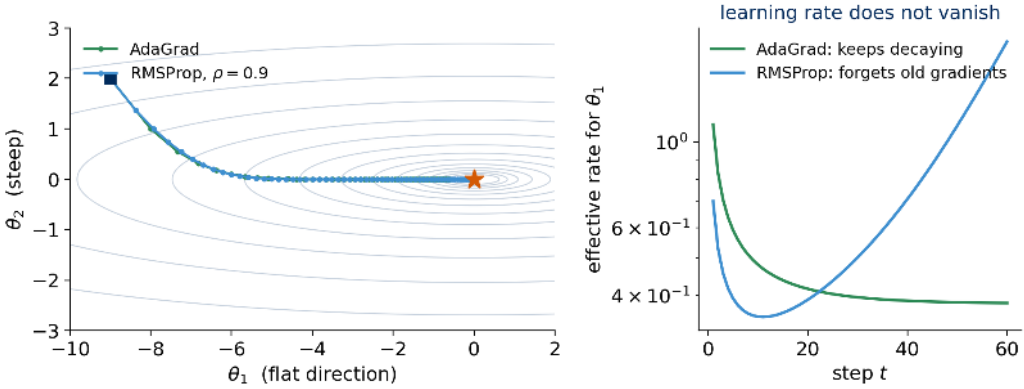
Replace the sum by an **exponentially weighted moving average**:

$$\boldsymbol{r}_{t+1} = \rho\, \boldsymbol{r}_t + (1-\rho)\, \boldsymbol{g}_t \odot \boldsymbol{g}_t, \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\boldsymbol{r}_{t+1}} + \epsilon} \odot \boldsymbol{g}_t$$

- old gradients forgotten at rate $\rho \approx 0.9$
- the learning rate no longer vanishes
- $\sqrt{\text{mean of } g^2}$ = root mean square of recent gradients — hence the name
- Adadelta ("adaptive delta): also replaces η by a running average of past updates

RMSProp Forgets Old Gradients

under the hood



left: AdaGrad and RMSProp ($\rho = 0.9$) on the toy loss $\frac{1}{2}(0.1\,\theta_1^2 + 4\,\theta_2^2)$; right: the effective learning rate of θ_1 at step t

Adam (Adaptive Moment Estimation): the Default Choice

under the hood

$$\boldsymbol{m}_{t+1} = \beta_1 \boldsymbol{m}_t + (1-\beta_1)\,\boldsymbol{g}_t$$
$$\boldsymbol{v}_{t+1} = \beta_2 \boldsymbol{v}_t + (1-\beta_2)\,\boldsymbol{g}_t^2$$
$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\,\frac{\hat{\boldsymbol{m}}}{\sqrt{\hat{\boldsymbol{v}}}+\epsilon}$$

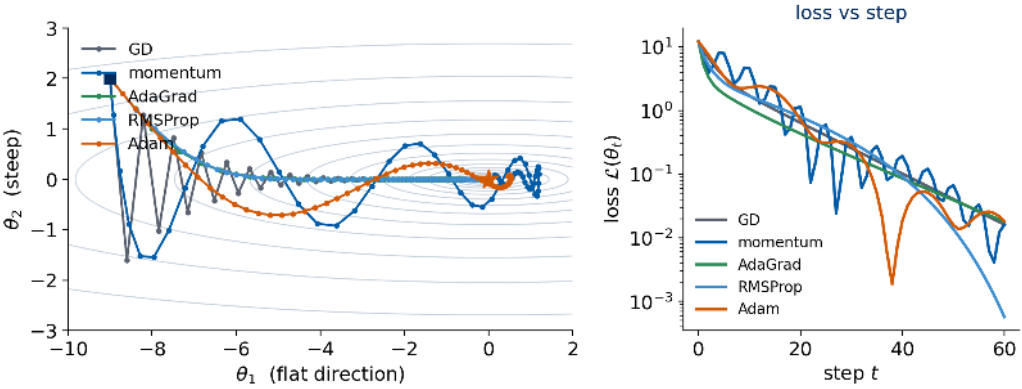
$\hat{\boldsymbol{m}} = \boldsymbol{m}_{t+1}/(1-\beta_1^{t+1})$,
$\hat{\boldsymbol{v}} = \boldsymbol{v}_{t+1}/(1-\beta_2^{t+1})$

- $\boldsymbol{m}$: 1st moment (mean) of the gradient = momentum
  $\boldsymbol{v}$: 2nd moment = RMSProp scaling
- typical: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\eta = 10^{-3}$

Adam: Kingma & Ba, arXiv:1412.6980; RMSProp: G. Hinton, Coursera lecture 6e (2012); Adadelta: M. Zeiler, arXiv:1212.5701

Five Optimisers on a Toy Loss

under the hood



GD = gradient descent; toy loss $\frac{1}{2}(0.1\,\theta_1^2 + 4\,\theta_2^2)$, 60 steps; on such a simple loss the ranking depends on the tuning — Adam's strength is robustness on real networks

6

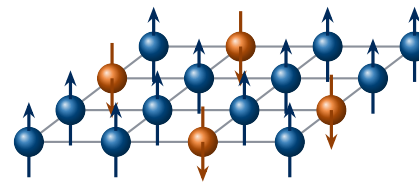
A Physics Example: the 2D Ising Model



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Classical Spins on a Lattice

- a square lattice $L \times L$; on every site i a *spin* $s_i = +1$ (up) or $s_i = -1$ (down)
- a simple model of a magnet: each spin is a tiny magnetic moment
- $2^{L \cdot L}$ possible configurations $\{s_i\}$



$s_i = +1$ $\quad$ $s_i = -1$

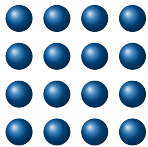
Energy of a configuration (Hamiltonian)

$$H(\{s_i\}) = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

$\langle i,j \rangle$: pairs of nearest neighbours; $J > 0$: aligned neighbours lower the energy (ferromagnet), we set $J = 1$; h: external magnetic field, we set $h = 0$.

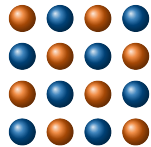
The Couplings Decide the Lowest-Energy State

$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$: the sign of J decides which neighbours are favourable. Lowest-energy configurations at $h = 0$:



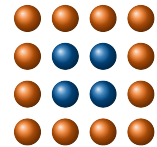
$J > 0$: ferromagnetic

neighbours *aligned*
all up or all down: 2 ground states
energy $-2J$ per spin
magnetisation $M = 1$



$J < 0$: antiferromagnetic

neighbours *opposite*
checkerboard pattern: 2 ground states
energy $-2|J|$ per spin
magnetisation $M = 0$



$J = 0$: non-interacting

spins independent
all 2^N configurations have the same energy
no order at any temperature

blue: $s_i = +1$, orange: $s_i = -1$.

Spin-Flip Symmetry and the External Field

$$H(\{s_i\}) = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

- square lattice: each spin has 4 neighbours, so 2 bonds per spin; hence the ground-state energy $-2|J|$ per spin
- $h = 0$: flipping *all* spins leaves H unchanged — hence two equivalent ground states
- $h > 0$: spins prefer $+1$; $h < 0$: spins prefer -1; the field removes this symmetry and selects one state

From here on: $J = 1$ (ferromagnet), $h = 0$.

The State at Finite Temperature

$T = 0$: one configuration

the system sits in a configuration of minimal energy: all spins up, or all spins down

$T > 0$: a statistical ensemble

the state is *not* a single configuration, but a probability for *every* configuration (Gibbs state, $k_B = 1$)

$$p(\{s_i\}) = \frac{e^{-H(\{s_i\})/T}}{Z}, \qquad Z = \sum_{\{s_i\}} e^{-H(\{s_i\})/T}$$

Z is the **partition function**: the normalisation of the probabilities.

Temperature sets how much weight high-energy configurations receive.

Two Spins at Finite Temperature

Example: two spins, $H = -Js_1 s_2$, four configurations, $Z = 2e^{J/T} + 2e^{-J/T} = 4\cosh(J/T)$:

configuration	**energy**	**probability**	$T \to 0$	$T \to \infty$
$\uparrow\uparrow,\ \downarrow\downarrow$	$-J$	$e^{+J/T}/Z$ each	$1/2$ each	$1/4$ each
$\uparrow\downarrow,\ \downarrow\uparrow$	$+J$	$e^{-J/T}/Z$ each	0	$1/4$ each

An observable is an average over this distribution, for example

$$\langle s_1 s_2 \rangle = \sum_{\{s_i\}} s_1 s_2 \, p(\{s_i\}) = \tanh(J/T),$$

equal to 1 at $T \to 0$ (aligned), 0 at $T \to \infty$ (independent).

Thermal Equilibrium: Energy against Disorder

The same rule for a lattice of $N = L^2$ spins and 2^N configurations:

$$p(\{s_i\}) = \frac{e^{-H(\{s_i\})/T}}{Z}, \qquad Z = \sum_{\{s_i\}} e^{-H(\{s_i\})/T}$$

Low temperature

low-energy configurations dominate: neighbours align, **ordered phase**

High temperature

all configurations become equally likely: spins point at random, **disordered phase**

Two observables of a configuration ($N = L^2$ spins):

$$M(\{s_i\}) = \frac{1}{N}\Big|\sum_i s_i\Big| \quad \text{(magnetisation)}, \qquad E(\{s_i\}) = -\frac{1}{N}\sum_{\langle i,j\rangle} s_i s_j \quad \text{(energy per spin)}$$

Thermal averages: $\langle M\rangle = \sum_{\{s_i\}} M(\{s_i\})\, p(\{s_i\})$, and the same for $\langle E\rangle$.

From the Partition Function to Observables

$Z(\beta, h) = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}$, with $\beta = 1/T$ and $H = -J \sum_{\langle i,j \rangle} s_i s_j - h\, S$, where $S = \sum_i s_i$ is the total magnetisation.

Derivative with respect to β: energy

$$\frac{\partial \ln Z}{\partial \beta} = \frac{1}{Z} \sum_{\{s_i\}} (-H)\, e^{-\beta H} = -\langle H \rangle$$

$$\frac{\partial^2 \ln Z}{\partial \beta^2} = \langle H^2 \rangle - \langle H \rangle^2$$

Derivative with respect to h: magnetisation

$$\frac{\partial \ln Z}{\partial h} = \frac{1}{Z} \sum_{\{s_i\}} \beta S\, e^{-\beta H} = \beta \langle S \rangle$$

$$\frac{\partial^2 \ln Z}{\partial h^2} = \beta^2 (\langle S^2 \rangle - \langle S \rangle^2)$$

The step used twice: $\partial_x \ln Z = \frac{1}{Z} \partial_x Z$, and $\frac{1}{Z} e^{-\beta H}$ is the Boltzmann probability.

Observables as Derivatives of $\ln Z$

The derivatives of $\ln Z(\beta, h)$, with $\beta = 1/T$ and total magnetisation $S = \sum_i s_i$:

observable	**from** $\ln Z$	**meaning**
mean energy	$\langle H \rangle = -\partial_\beta \ln Z$	first derivative: an average
mean magnetisation	$\langle S \rangle = \frac{1}{\beta}\, \partial_h \ln Z$	first derivative: an average
specific heat	$C = \dfrac{d\langle H \rangle}{dT} = \beta^2\, \partial_\beta^2 \ln Z$	second derivative: energy fluctuations
susceptibility	$\chi = \dfrac{d\langle S \rangle}{dh} = \frac{1}{\beta}\, \partial_h^2 \ln Z$	second derivative: magnetisation fluctuations

Per spin: divide by N.

One function, $\ln Z(\beta, h)$, contains all equilibrium observables of the model.

The 1D Ising Chain

N spins on a line, $H = -J\sum_i s_i s_{i+1}$, $h = 0$. Exact solution (Ising, 1925), ring of N spins: $Z = \left(2\cosh(J/T)\right)^N + \left(2\sinh(J/T)\right)^N$, so $\frac{1}{N}\ln Z \to \ln\left(2\cosh(J/T)\right)$ for $N \to \infty$, smooth in T for every $T > 0$.

From $\ln Z$ and the transfer matrix:

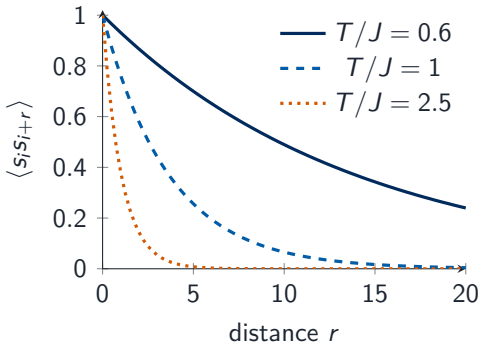
- energy per spin: $-J\tanh(J/T)$
- specific heat: $(J/T)^2/\cosh^2(J/T)$ (smooth, no peak divergence)
- **correlations:** $\langle s_i s_{i+r}\rangle = \tanh^r(J/T) = e^{-r/\xi}$
- **correlation length:** $\xi = -1/\ln\tanh(J/T)$

ξ is finite at every $T > 0$: no phase transition.

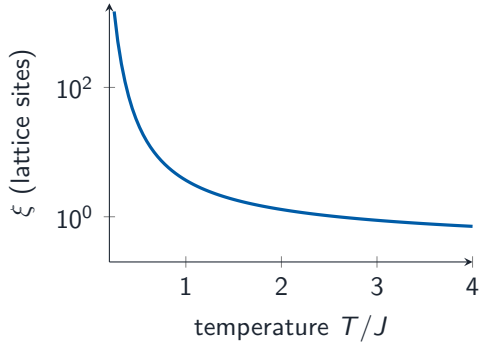
E. Ising, *Beitrag zur Theorie des Ferromagnetismus*, Z. Phys. **31**, 253 (1925), doi:10.1007/BF02980577

Correlations in the 1D Chain Decay Exponentially

Exact results for the infinite chain: $\langle s_i s_{i+r} \rangle = \tanh^r(J/T)$ and $\xi = -1/\ln\tanh(J/T)$.



exponential decay at every T



$\xi \approx \frac{1}{2} e^{2J/T} \to \infty$ only as $T \to 0$

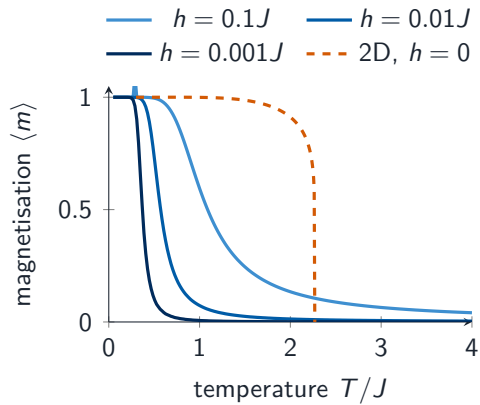
1D: Magnetisation as a Function of Temperature

Switch on a small field h to pick a direction:

$$\langle m \rangle = \frac{\sinh(h/T)}{\sqrt{\sinh^2(h/T) + e^{-4J/T}}}$$

- in a field, spins align at low T
- make the field weaker: the curve shifts to $T \to 0$
- $h \to 0$: $\langle m \rangle = 0$ at *every* $T > 0$

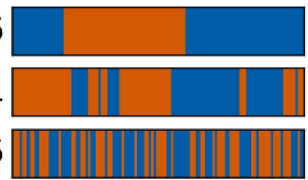
Contrast with 2D (Onsager): $\langle m \rangle \neq 0$ below $T_c \approx 2.269\,J$ even at $h = 0$ — **spontaneous magnetisation**.



solid: 1D chain in a field h; dashed: 2D at $h = 0$ (Onsager)

$\langle m \rangle = 0$ Does Not Mean Random Spins

Exact samples of a 1D chain ($N = 120$, $h = 0$), blue $= +1$, orange $= -1$:



- all three have $m \approx 0$, but only the high-T chain is random
- low T: **long ordered domains**, up and down in equal measure — locally ordered, globally $m \approx 0$
- $\langle m \rangle$ is an average; the **correlation length** ξ measures how ordered the spins are

Domain Walls in 1D at Every Temperature

The system minimises the free energy $F = E - TS$: energy against entropy.

- one domain wall costs $\Delta E = 2J$ (one broken bond)
- but it can sit at any of $N - 1$ places: $\Delta S = \ln(N-1)$

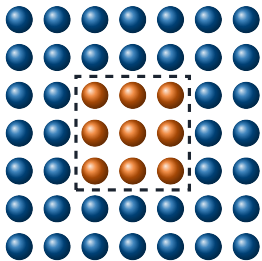
$$\Delta F = 2J - T\ln(N-1) < 0 \quad \text{for a long chain at } \textit{any } T > 0$$

Walls proliferate until they are $\sim \xi$ apart: domains in 1D at every $T > 0$.

Two dimensions

In 2D a wall is a line — its cost grows with its length (next slide).

2D: Domain Walls



a flipped island; its boundary is a domain wall of length ℓ (here $\ell = 12$)

- energy cost grows with the wall: $\Delta E = 2J\ell$
- number of wall shapes of length ℓ: at most about 3^ℓ (three directions at every step), so $\Delta S \approx \ell \ln 3$
- $\Delta F \approx \ell\,(2J - T \ln 3)$: positive at low temperature, large islands are suppressed

Order Survives at Low Temperature in 2D

Free-energy cost of a wall of length ℓ:

$$\Delta F \approx \ell\,(2J - T \ln 3)$$

- **order survives** below a temperature of the order of $2J/\ln 3 \approx 1.8\,J$ — a rough estimate; the exact value is $T_c \approx 2.269\,J$

Dimension decides

1D: wall energy is constant, entropy wins at every $T > 0$.
2D: wall energy grows with its length, order wins at low T.

R. Peierls, *On Ising's model of ferromagnetism*, Proc. Camb. Phil. Soc. **32**, 477 (1936), doi:10.1017/S0305004100019174

2D: Onsager's Solution of the Partition Function

Every observable is a derivative of $\ln Z$, but $Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}$ is a sum over 2^N configurations. **Computing $\ln Z$ in closed form solves the model.**

Onsager (1944): $\ln Z$ computed exactly for the 2D square lattice, $h = 0$, $N \to \infty$

$$\frac{1}{N}\ln Z = \ln\bigl(2\cosh(2\beta J)\bigr) + \frac{1}{2\pi}\int_0^{\pi} d\phi\, \ln\Bigl[\tfrac{1}{2}\Bigl(1+\sqrt{1-\kappa^2\sin^2\phi}\Bigr)\Bigr], \qquad \kappa = \frac{2\sinh(2\beta J)}{\cosh^2(2\beta J)}$$

- $\kappa \le 1$ for every temperature, and $\kappa = 1$ exactly when $\sinh(2\beta J) = 1$: there the square root vanishes at $\phi = \pi/2$ and $\ln Z$ is **not analytic**
- $\sinh(2J/T_c) = 1 \;\Rightarrow\; T_c = \dfrac{2J}{\ln(1+\sqrt{2})} \approx 2.269\, J$

L. Onsager, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117

Specific Heat and Spontaneous Magnetisation in 2D

- at T_c the specific heat C diverges logarithmically: a second-order (continuous) phase transition
- the spontaneous magnetisation

$$\langle M \rangle = \left[1 - \sinh^{-4}(2J/T)\right]^{1/8}, \qquad T < T_c,$$

was announced by Onsager in 1949; the derivation was published by C. N. Yang in 1952

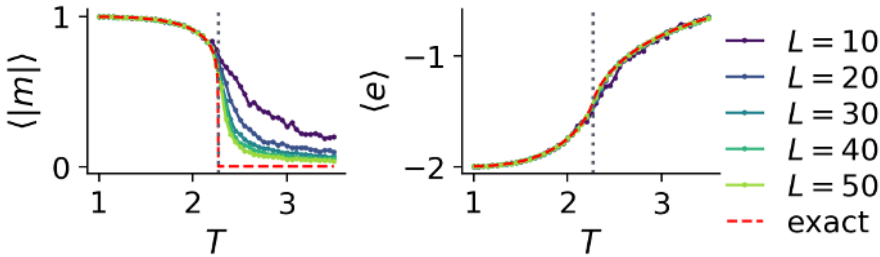
L. Onsager, *Crystal Statistics. I. A Two-Dimensional Model with an Order-Disorder Transition*, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117; C. N. Yang, *The Spontaneous Magnetization of a Two-Dimensional Ising Model*, Phys. Rev. **85**, 808 (1952), doi:10.1103/PhysRev.85.808

A Phase Transition at a Critical Temperature

- **1D** (Ising, 1925): no phase transition at any $T > 0$
- **2D** (Onsager, 1944, exact solution at $h = 0$): a transition at $T_c = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$
- $T < T_c$: spontaneous magnetisation without external field, $\langle |m| \rangle = \left[1 - \sinh^{-4}(2/T)\right]^{1/8}$ for $L \to \infty$
- $T > T_c$: $\langle |m| \rangle = 0$ for $L \to \infty$

$m = \frac{1}{N} \sum_i s_i$ is the **order parameter**: a quantity that tells the two phases apart. Near T_c: $\langle |m| \rangle \sim (T_c - T)^{1/8}$, and fluctuations of observables become large.

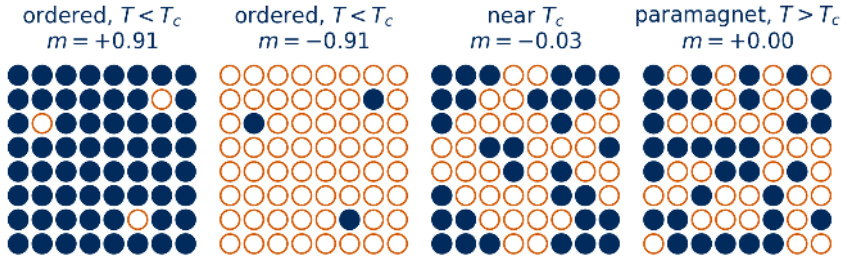
Magnetisation and Energy from Monte Carlo



$m = \frac{1}{N} \sum_i s_i$, $e = E/N$; solid: Monte Carlo, $L \times L$ lattices; dashed red: exact $L \to \infty$ (Onsager, Yang); dotted: T_c.

On a finite lattice $\langle |m| \rangle > 0$ also above T_c; the curves approach the exact result as L grows.

Ordered and Disordered Phases



order parameter: magnetisation per spin $m = \frac{1}{N} \sum_i s_i$

	ordered, $T < T_c$	**disordered**, $T > T_c$
spins	mostly parallel, all up or all down	random, small domains
m	$\pm m_0(T) \neq 0$: symmetry $s_i \to -s_i$ (all i) broken	$m \to 0$ for $L \to \infty$

Reversing the Magnetisation Costs Two Walls

By symmetry $\langle m \rangle = 0$ on a finite lattice, yet below T_c each configuration has $m \approx +m_0$ or $m \approx -m_0$.

$$\sigma(T) = 2J + T \ln \tanh \frac{J}{T} \qquad \text{(interface tension, Onsager)}$$

- σ: free energy per unit length of a straight wall; $\sigma(0) = 2J$, $\sigma(T_c) = 0$
- from $+m_0$ to $-m_0$: a flipped band with **two walls of length L**, relative weight $\sim e^{-2\sigma L/T}$
- $T = 2.0$: $\sigma = 0.456$; weight 10^{-2} for $L = 10$, 10^{-10} for $L = 50$

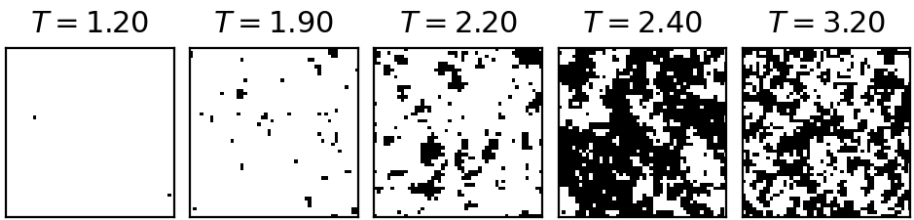
For $L \to \infty$ the chosen sign is kept: spontaneous symmetry breaking. Hence we measure $|m|$.

L. Onsager, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117; K. Binder, Phys. Rev. A **25**, 1699 (1982), doi:10.1103/PhysRevA.25.1699

A Continuous Transition at $T_c = 2.269$

$T_c = 2/\ln(1+\sqrt{2}) = 2.269$ (Onsager); reduced temperature $t = (T - T_c)/T_c$

- **continuous** (second order): m vanishes without a jump, $m \sim (-t)^{\beta_M}$, $\beta_M = 1/8$; at a **first-order** transition the order parameter jumps



Monte Carlo configurations, $L = 50$, across T_c

Critical Exponents of the 2D Ising Model

Near T_c, with reduced temperature $t = (T - T_c)/T_c$:

quantity	behaviour	exponent
order parameter m $(T < T_c)$	$m \sim (-t)^{\beta_M}$	$\beta_M = 1/8$
correlation length ξ	$\xi \sim \lvert t \rvert^{-\nu}$	$\nu = 1$
susceptibility χ	$\chi \sim \lvert t \rvert^{-\gamma}$	$\gamma = 7/4$
specific heat c	$c \sim -\ln \lvert t \rvert$	logarithmic

The exponents are shared by all models with the same symmetry and dimension: the **2D Ising universality class**.

L. Onsager, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117; C. N. Yang, Phys. Rev. **85**, 808 (1952), doi:10.1103/PhysRev.85.808

Mean Field Neglects Fluctuations

Mean field: each spin feels its z neighbours only through their average m, i.e. $\langle s_i s_j \rangle \to \langle s_i \rangle \langle s_j \rangle$:

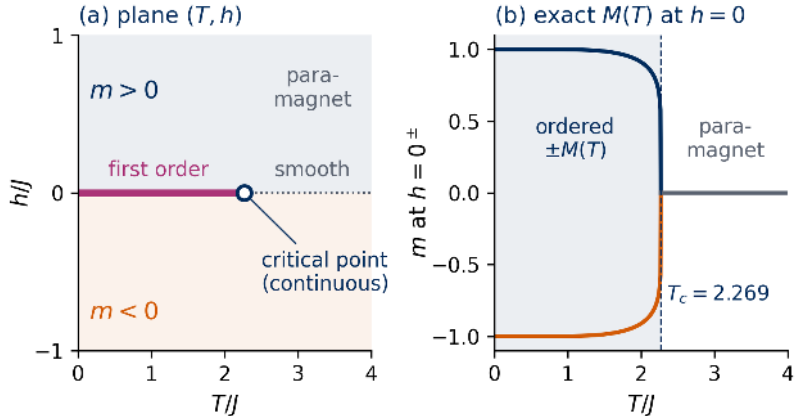
$$m = \tanh\!\left(\frac{zJm}{T}\right) \quad \Rightarrow \quad m \neq 0 \text{ for } T < T_c^{\mathrm{MF}} = zJ$$

	mean field	**exact**
2D T_c ($z = 4$)	$4J$	$2.269\,J$
β_M	$1/2$	$1/8$
c at T_c	jump to $3/2$	$-\ln\lvert t\rvert$
1D ($z = 2$)	$T_c = 2J$	no transition

- near T_c correlated domains of size ξ dominate; in 2D they lower T_c by a factor 1.76
- mean-field exponents are exact only above $d = 4$

derivations and checks: Lab 012; M. Kardar, *Statistical Physics of Fields*, Cambridge University Press (2007), doi:10.1017/CBO9780511815881

Phase Diagram in the (T, h) Plane

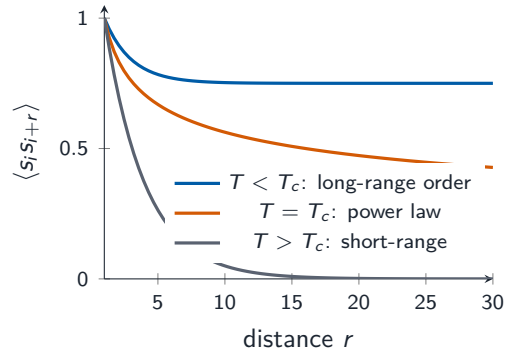


field h: $H = -J\sum_{\langle i,j\rangle} s_i s_j - h\sum_i s_i$; $h = 0$, $T < T_c$: **first-order** line, m jumps from $-m_0$ to $+m_0$; it ends at the **critical point** $(T_c, 0)$; no transition for $T > T_c$. The data sets of this lecture: $h = 0$, a scan in T through the critical point.

Criticality: Three Kinds of Correlations

Correlation of two spins at distance r: $\langle s_i s_{i+r} \rangle$. **Correlation length $\xi(T)$:** the distance over which spins are correlated (roughly the size of a domain).

- $T > T_c$: **short-range correlations**
  $\langle s_i s_{i+r} \rangle \sim e^{-r/\xi} \to 0$: exponential decay, distant spins are independent
- $T < T_c$: **long-range order**
  $\langle s_i s_{i+r} \rangle \to M^2 > 0$: spins stay correlated at any distance
- $T = T_c$: **critical (power-law) correlations**
  $\langle s_i s_{i+r} \rangle \sim r^{-1/4}$: no characteristic scale; $\xi \propto |T - T_c|^{-1}$ diverges



schematic curves (not simulation data)

Criticality: Fluctuations of Observables

The second derivatives of $\ln Z$ are *variances*: how much an observable changes from one configuration to the next.

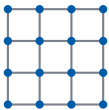
$$\langle H^2 \rangle - \langle H \rangle^2 = T^2\, C, \qquad \langle S^2 \rangle - \langle S \rangle^2 = T\, \chi$$

- far from T_c: small fluctuations; every sampled configuration looks like the others
- at T_c the response functions diverge: $\chi \sim |T - T_c|^{-7/4}$, $C \sim -\ln |T - T_c|$
- so the **fluctuations of magnetisation and energy are largest at T_c**: configurations at the *same* temperature differ strongly from each other

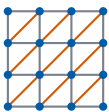
Consequence for learning

near T_c a single configuration does not reveal its phase reliably

T_c Depends on the Lattice, the Exponents Do Not



square, $z = 4$



triangular, $z = 6$

lattice	T_c/J	η from strips
square	$2/\ln(1+\sqrt{2}) = 2.269$	0.2507
triangular	$4/\ln 3 = 3.641$	0.2500

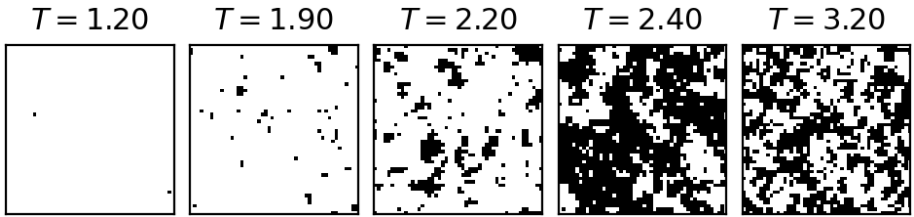
- η: $\langle s_0 s_r \rangle \sim r^{-\eta}$ at T_c; exact $\eta = 1/4$ on both lattices
- more neighbours keep the order longer: T_c is a lattice detail
- at T_c, $\xi \gg$ lattice spacing: the lattice details drop out of the exponents, as the **2D Ising universality class** requires

η: transfer matrices of infinite strips (Lab 012); square: width 12 at T_c; triangular: crossing of widths 10 and 12

R. M. F. Houtappel, Physica **16**, 425 (1950), doi:10.1016/0031-8914(50)90130-3

Sampling Spin Configurations

Goal: configurations drawn from $p(\{s_i\}) \propto e^{-H/T}$ at a chosen temperature T, without summing over 2^N configurations.



sampled configurations, 50×50 lattice, at $T = 1.2,\ 1.9,\ 2.2,\ 2.4,\ 3.2$ (black and white: the two spin values)

Low T: spins aligned. High T: a random pattern. Each image is one sample from the Boltzmann distribution.

The Metropolis Algorithm

Metropolis Monte Carlo for the 2D Ising model

1. initialise an $L \times L$ lattice with random spins $s_{i,j} = \pm 1$; set the temperature T
2. pick a site (i, j) and compute the energy change of flipping its spin:

$$\Delta E = 2\, s_{i,j} \left(s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1} \right)$$

3. if $\Delta E \leq 0$: flip the spin, $s_{i,j} \leftarrow -s_{i,j}$
4. if $\Delta E > 0$: draw a random number $r \in [0, 1)$ and flip the spin if $r < e^{-\Delta E / T}$

Only the four nearest neighbours enter ΔE.

Monte Carlo Steps and Measurements

Steps 3 and 4 accept a proposed flip with probability

$$P_{\text{acc}} = \min\left(1,\, e^{-\Delta E/T}\right) :$$

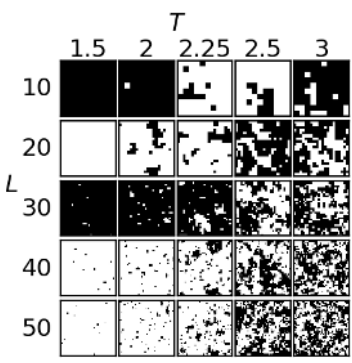
energetically unfavourable flips are accepted more often at high T.

Metropolis Monte Carlo for the 2D Ising model

5. repeat for all sites (one Monte Carlo step); repeat many steps
6. after equilibration: store configurations and measure M, E

N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, E. Teller, *Equation of State Calculations by Fast Computing Machines*, J. Chem. Phys. **21**, 1087 (1953), doi:10.1063/1.1699114

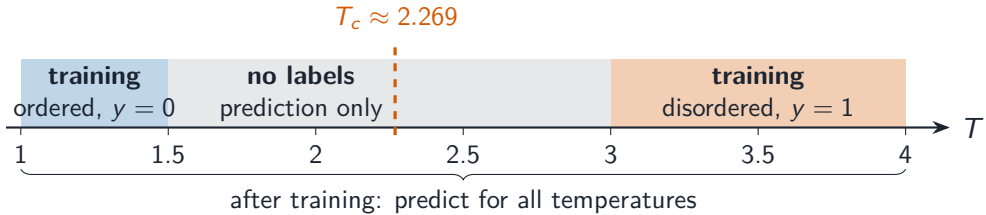
The Dataset



- rows: lattice size $L = 10, \ldots, 50$; columns: temperature T
- Metropolis sampling, 51 temperatures $T = 1.0, 1.05, \ldots, 3.5$, 200 configurations per temperature and size
- each configuration is a black-and-white $L \times L$ image
- low T: almost all spins aligned (all up or all down); high T: random pattern
- $T = 2.25$ and $T = 2.5$ look alike, but lie on **opposite sides** of $T_c \approx 2.269$

The task for a network: find T_c from the images alone, without being told about magnetisation.

Supervised Learning Far from the Transition



1. **Data:** X = spin configuration, y = phase label

2. **Train only where the answer is clear:**
   $T < T_{\text{min}} = 1.5$: ordered, $y = 0$; $T > T_{\text{max}} = 3$: disordered, $y = 1$
   configurations with $T_{\text{min}} \leq T \leq T_{\text{max}}$ are *excluded* from training

The Network and the Prediction Confidence

3. **Model:** a small CNN — two blocks of Conv2D (16 kernels 3×3, periodic boundaries) + ReLU + 2×2 MaxPooling, global average pooling, dense 32 (ReLU), 2 outputs (softmax); cross-entropy, Adam optimiser. Global average pooling: one network for every L

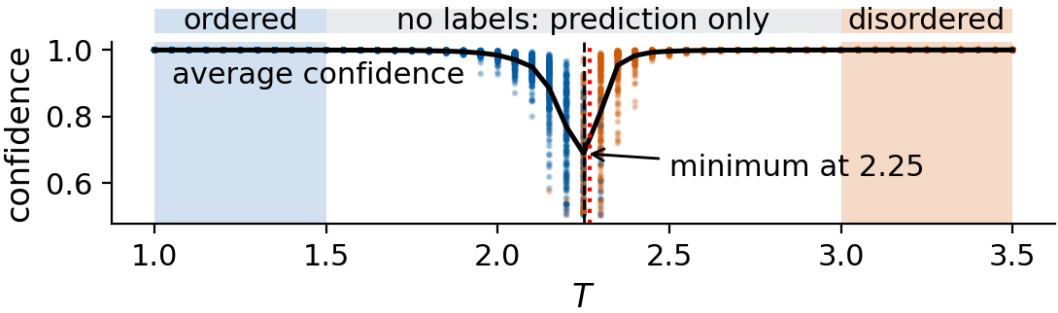
4. **Predict** on the whole temperature range, including the region the network has never seen

5. **Plot the prediction confidence** against T

Prediction confidence

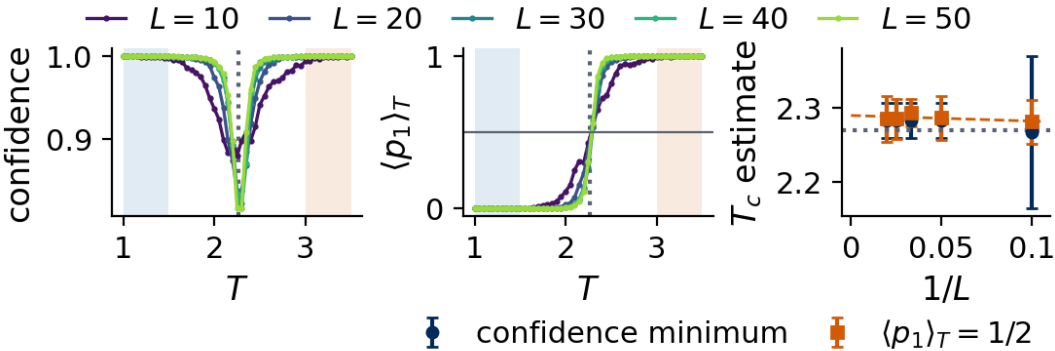
the softmax probability of the predicted class: for an output $q(x) = [0.1, 0.9]$ the predicted class is 1 and the confidence is 0.9

The Confidence Drops at the Critical Temperature



confidence $= \max(p_0, p_1)$, p_1 = network probability of "disordered"; dots: single configurations (blue: predicted ordered, orange: disordered); black: average over the 200 configurations at each T. $L = 50$: lowest average confidence at $T = 2.25$ (temperature step 0.05); exact $T_c \approx 2.269$ (red dotted). Near T_c the magnetisation vanishes and fluctuations are large: the two phases are hardest to tell apart.

The Estimate of T_c for Lattice Sizes 10 to 50



left: average confidence $\langle \max(p_0, p_1) \rangle_T$; middle: $\langle p_1 \rangle_T$, average over the configurations at T; right: two estimates of T_c, mean $\pm$ standard deviation over 3 trainings (seeds); dashed: linear fit of the crossing in $1/L$

Mean Estimates Lie within 0.025 of the Exact T_c

- the same network and training windows for every L; larger L: a narrower, deeper dip and a steeper $\langle p_1 \rangle_T$
- crossing $\langle p_1 \rangle_T = 1/2$: 2.281, 2.286, 2.293, 2.285, 2.285 for $L = 10, \ldots, 50$; confidence minimum: 2.267 for $L = 10$, 2.283 for $L \geq 20$
- every mean estimate lies within 0.025 of $T_c = 2.269$, one half of the temperature step 0.05; the offsets show no trend with L and are comparable to the seed-to-seed standard deviation (0.02–0.03; 0.10 for the confidence minimum at $L = 10$)

data and trainings: notebook Ising_2D_Monte_Carlo_and_CNN; Lab 011 repeats the study with a PyTorch network for $L = 10, 20, 40$

Gains and Limits of the CNN Estimate of T_c

Gains

- an estimate of T_c from raw configurations, with no formula for the order parameter
- for many systems of current interest the order parameter is *not known*: the network signals that *something happens* near a given temperature

Limits

- finite lattice: the transition is smoothed and the dip is broadened; larger L narrows the dip
- the labels and the two training windows came from us
- the network does not tell us *why* there is a transition

The network reproduces a known T_c; it does not explain the transition.

The Chinese Room Thought Experiment

1. A person who does not understand Chinese sits inside a room.
2. The person has a rulebook (an algorithm) that says how to manipulate Chinese characters to produce correct responses to questions written in Chinese.
3. To an observer outside, the room appears to understand Chinese: the answers are coherent.
4. The person has no understanding of the language — they are only following rules.



J. R. Searle, *Minds, brains, and programs*, Behav. Brain Sci. **3**, 417 (1980), doi:10.1017/S0140525X00005756

The Chinese Room and Deep Learning

- **Deep learning is data manipulation:** a network processes inputs and generates outputs by recognising patterns and applying transformations, as the person in the room follows rules
- **Statistical, not semantic:** the network's knowledge is a set of correlations in the data; it does not "know" what the symbols (words, images, spins) mean
- **Symbolic vs subsymbolic:** the room works with explicit symbolic rules; a network uses subsymbolic patterns (weights, activations) derived from data
- **Simulation, not cognition:** plausible outputs based on training data

Back to the Ising model

The CNN located T_c, but it gave "some solution" without an explanation: no order parameter, no mechanism of the transition. The understanding — Onsager's solution, the order parameter — remains the scientist's job.

7

Wrap-up



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Summary

- A neural network is a non-linear, parametrised function with nested structure
- Elements of deep learning: data $\rightarrow$ loss function $\rightarrow$ architecture $\rightarrow$ training algorithm
- Classification: cross-entropy from the Kullback–Leibler divergence
- CIFAR-10: dense networks vs CNNs; pooling; overfitting and regularisation
- Class activation maps: where the CNN looks; adversarial examples: where it fails
- Optimisers: SGD, momentum, AdaGrad, RMSProp, Adam
- 2D Ising model: a network trained far from T_c locates the transition
- Spin glasses: frustration, frozen disorder; Hopfield network: memories as energy minima, capacity $\approx 0.14N$

Thank you!

Questions?